

TAMPINES JUNIOR COLLEGE PRELIMINARY EXAMINATION 2008

BIOLOGY Paper 2 Core Paper

9747/02

QUESTION ONE

(a)	Secondary structure – alpha helices, beta-pleated sheets;	1
	Tertiary structure – 3-dimensional, globular shape;	1
	Quaternary structure - 2 subunits / 2 polypeptide chains;	1
(b)	(i) microtubule organising centre / MTOC; (also accept centrosome/ centriole/ nuclear envelope/ kinetochore)	1
	(ii) they are spindle fibres for separating sister chromatids / chromosomes / kinetochores;	1
(c)	The chromosomes attached to microtubules at centromeres would line up on the equator of the spindle and at right angles to the spindle axis;	1
	Centromeres of each chromosome divides;	1
	Sister chromatids separate and sister chromatids become daughter chromosomes;	1
	Daughter chromosomes move to opposite poles of the spindle as spindle fibres shorten so that the spindle fibres/ kinetochore attached to microtubules pull daughter centromeres and daughter chromosomes apart;	1
		Max. 3
(d)	Chromosomes uncoiling, lengthen and cannot be seen clearly;	1
	Nuclear envelope reforming;	1
	Nucleoli reappearing;	1
	Spindle microtubules dispersing;	1
	Cytokinesis – cell plate formed at the position of formerly the equator of the spindle and new cell walls formed	1
		Max. 3
(e)	The exact replication of DNA from the parental chromosomes before mitosis occurs;	1
	The arrangement of the chromosomes on the spindle ensures that the sister chromatids are distributed evenly between the 2 daughter cells;	1
	Mitosis ensures that the daughter cells receive precisely the same number and types of chromosomes as the original parent cell;	1
	No chiasmata formation and therefore no crossing over;	1
		Max. 3
	Total	Max. 14



QUESTION TWO

(a)	(i) transfer RNA	1
	(ii) hydrogen bond	1
(b)	codon	1
(c)	TTA	1
(d)	Endoplasmic reticulum – transports the protein and modification of protein – folding to secondary / tertiary structure	1
	Golgi apparatus – transport and modification of protein eg. Addition of COH	1
(e)	Cellulase Structure – globular Function – enzymatic	1 1
	Collagen Structure - fibrous	1
	Total	Max. 9



QUESTION THREE

(a)	1. Nucleic acid in viruses is either DNA or RNA whereas prokaryotes has both DNA and RNA.	Any 3 differences, no comparison with prokaryotes (1/2 only)
	2. nucleic acid can be linear or circular in viruses, it is circular in prokaryotes	
	3. Absence of ribosomes in viruses whereas prokaryotes has ribosomes	
	4. Only protein coat present in viruses whereas prokaryotes has a cell surface membrane, cell wall and capsule/ slime layer.	
	5. Absence of glycoproteins on prokaryotes cell surface but glycoproteins are present on the virus envelope.	
	6. viruses are smaller in size than prokaryotes	
	7. no flagella on viruses, present on prokaryotes	
(b)	Lacking other structures/ organelles required for production of proteins and replication of nucleic acids;	1
	Enters cell and makes use of host's replicative mechanisms/protein synthesis machinery to synthesis its genome and proteins	1
	For replication/forming more copies of itself	1
(c)	(i) 2 000 amino acids	1
	(ii) 3 nitrogenous bases (constituting one codon) codes for one amino acid;	1
	Hence a single strand of RNA with 6 000 nucleotides would code for 6 000 divided by 3 = 2 000 amino acids	1
(d)	(i) lysis of the bacterial cell wall;	1
	(ii) Involvement of glycoproteins on the influenza viral envelope binding with receptors on the host cell membrane;	1
	Endocytosis occurs; fusion of envelope and cell membrane to release genome into cell at lowered pH	1
	Total	Max. 12



QUESTION FOUR

(a) (i)	kinase catalyses the phosphorylation of Rb protein;	1
	Rb protein binds to gene regulatory proteins and prevents them from activating cell proliferation;	1
(ii)	mutation of Rb gene involves loss of Rb protein and hence loss of control of cell cycle → loss of function of gene → not a proto-oncogene (gain of function	1
(b) (i)	chromosome 21;	1
	Times 3 / trisomy/ extra copy	1
	Translocation of fragment of chromosome 21	1
	On to another chromosome	1
		Max 3
(ii)	3 copies for a particular chromosome / homologous pair + 1 / $2n + 1$	1
(iii)	homologous chromosomes fail to separate / nondisjunction;	1
	At metaphase I / anaphase I / meiosis I;	1
	In oogenesis;	1
	Secondary oocyte has both members of homologous pair of chromosomes;;	1
	Third chromosome added;	1
	(from sperm) at fertilisation;	1
	(A) converse nondisjunction in spermatogenesis;	
		Max. 3
(iv)	Both have, three / an extra copy of Ets 2;	1
	Both produce more Ets 2 messenger RNA than normal;	1
	Both have more transcription factor than normal;	1
		Max. 2
	Total	12



QUESTION FIVE

(a)	Expected phenotypes: Black, brown, golden	1																									
	Proportions: 9 black: 3 brown: 4 golden	1																									
	Derivation: Parental Phenotypes: Heterozygous Black Heterozygous Black Parental Genotypes: BbEe X BbEe	1																									
	Gametes: BE Be bE be for both	1																									
	Punnett square: <table><tr><td></td><td>BE</td><td>Be</td><td>bE</td><td>be</td></tr><tr><td>BE</td><td>BBEE</td><td>BBEe</td><td>BbEE</td><td>BbEe</td></tr><tr><td>Be</td><td>BBEe</td><td>BBe^e</td><td>BbEe</td><td>Bbe^e</td></tr><tr><td>bE</td><td>BbEE</td><td>BbEe</td><td>bbEE</td><td>bbEe</td></tr><tr><td>be</td><td>BbEe</td><td>Bbee</td><td>bbEe</td><td>bbee</td></tr></table> Black (9) Brown (3) Golden (4)		BE	Be	bE	be	BE	BBEE	BBEe	BbEE	BbEe	Be	BBEe	BBe ^e	BbEe	Bbe ^e	bE	BbEE	BbEe	bbEE	bbEe	be	BbEe	Bbee	bbEe	bbee	2
	BE	Be	bE	be																							
BE	BBEE	BBEe	BbEE	BbEe																							
Be	BBEe	BBe ^e	BbEe	Bbe ^e																							
bE	BbEE	BbEe	bbEE	bbEe																							
be	BbEe	Bbee	bbEe	bbee																							
(b)	Most Probable genotype of III-1: BbEe NB. bbEe X BbEe <table><tr><td></td><td>BE</td><td>Be</td><td>bE</td><td>be</td></tr><tr><td>bE</td><td>BbEE</td><td>BbEe</td><td>bbEE</td><td>bbEe</td></tr><tr><td>be</td><td>BbEe</td><td>Bbee</td><td>bbEe</td><td>bbee</td></tr></table>		BE	Be	bE	be	bE	BbEE	BbEe	bbEE	bbEe	be	BbEe	Bbee	bbEe	bbee	1										
	BE	Be	bE	be																							
bE	BbEE	BbEe	bbEE	bbEe																							
be	BbEe	Bbee	bbEe	bbee																							
(c)	Possible genotypes of III-7: Bbee and bbee	1																									
	Derivation: ‘ee’ as has a golden coat ;	1																									
	And cannot be ‘BB’ but must be ‘Bb’ or ‘bb’ as male parent is brown	1																									
	Total	Max. 10																									



QUESTION SIX

(a)	Oxygen needed;	1
	Positive correlation/ more oxygen more absorption;	1
	Aerobic respiration/ oxygen needed for respiration;	1
	Ref. to oxidative phosphorylation;	1
	Active transport involved;	1
	Only 2% oxygen needed;	1
	Little increase above 2%;	1
		Max. 3
(b)	Absorption reduced;	1
	Figures to support;	1
	Ref. to small change between 5 and 10;	1
	DNP allows electrons down the chain;	1
	No ATP produced by ETC;	1
	Therefore less ATP;	1
	Therefore less active transport;	1
		Max. 4
(c)	(i) Less absorption;	1
	(ii) Krebs cycle produces NADH;	1
	NADH supplies energy / H^+ / e^- ;	1
	For oxidative phosphorylation;	1
	Less energy / ATP for active transport;	1
		Max. 3
(d)	Information, storage / retrieval;	Any 2
	DNA / RNA;	
	Energy transfer;	
	ATP / ADP;	
	Sugar phosphates ;	
	Ref. to respiration / glycolysis;	
	Ref. to photosynthesis;	
	Secondary messenger;	
	Cyclic AMP;	
	Membranes;	
	Phospholipids;	
	Skeleton;	
	Bone/ teeth;	
	AVP ;	
Total		Max. 13



QUESTION SEVEN

(a)	Base substitution in the code for glutamic acid (in the gene coding for the β -globin chain of haemoglobin);	1
	caused by a mutagen, for example radioactivity;	1
	altered the codon, with thymine being replaced by adenine;	1
	Resulting in valine replacing glutamic acid at the 6 th amino acid of the β -globin chain and hence Haemoglobin A $\rightarrow$ Haemoglobin S;	1
	Gene mutation/ Point mutation;	
	Stating the respective triplets CTT/ CTC (glutamic acid) and CAT / CAC (valine);	
		Max 3
(b)	Conditions: low levels of oxygen;	1
	Abnormal beta chains make the haemoglobin molecules less soluble, tend to stick together, forming long fibres inside the red blood cells.	1
	Ref. to the change in structure and shape of the haemoglobin, mentioning the change in the bonding and the hydrophobic groups facing outwards;	1
	Glutamic acid is hydrophilic in nature, valine is hydrophobic in nature $\Rightarrow$ hence ionic cross link is not formed and hence there is a change in the shape of the tertiary structure of haemoglobin;	
	The red blood cells are pulled out of shape, into a crescent or sickle shape.	1
	Bonus mark allowed	Max. 3
(c)	Distribution same as / very similar to, the malarial parasite;	1
	Resistant to malaria;	1
	Some migration into Nairobi;	1
	Allele carried by, immigrants/ urban commuters;	1
	Similar to breeding grounds of the mosquito;	1
	Act as a vector/ carries malarial parasite;	1
	(sickle cell) not found above 1 500 m;	1
	No malaria/ no mosquitoes/ low oxygen;	1
		Max. 2
(d)	Heterozygote population / proportion of sickle cell allele / gene frequency falls;	1
	No longer of selective advantage to be resistant to malaria / normal homozygous genotype has no selective disadvantage	1
	Instead is now at a selective disadvantage to be suffering from anaemia / oxygen deprivation;	1
		Max. 3
	Total	Max. 10



QUESTION EIGHT

- (a) Describe the differences in structure between the prokaryotic and eukaryotic genome. [8]

	Prokaryotic cell	Eukaryotic cell
1	Supercoiling takes place to compact the genome, maintained by HU, H-NS	Coiling around histones;
2	50 DNA looped domains are held together by a protein scaffold. No mention of proteins required OR Mention of proteins only without description of organisation (1/2m)	forming nucleosomes/ which are then further organized into solenoids/ further condensed into a chromosome
3	Single, circular DNA (for most prokaryotes)	Two or more linear chromosomes
4	Usually one ORI	Usually multiple ORIs
5	Cluster of structural genes regulated by a promoter/ OR Regulated by operon system	Each gene has its own promoter, termination sequence and other control sequences/ OR NO operon system
6	Continuous OR Non-coding sequences Present, in very <u>small</u> amounts (usually <15% of total genome);	Interrupted by introns OR Non-coding sequences present, in <u>large</u> amounts.
7	functionally related genes located adjacent to one another and expressed as a single unit. Genes in operon are coding for a set of proteins in a single pathway via the translation of a <u>polycistronic</u> mRNA	Functionally related genes are expressed individually as single units/can be far away from each other Each mature <u>monocistronic</u> eukaryotic mRNA codes for one protein.
NOT MORE THAN 2 MARKS FROM THE FOLLOWING:		
8	Tandemly repeated genes not common in prokaryotes/ Contain much lesser repetitive DNA sequences	Tandemly repeated genes present/ Contains repetitive DNA sequences of varying lengths.
9	No telomeres in prokaryotic chromosomes – circular form	Contain telomeres, section of DNA which marks the end of the chromosome, rather than unnaturally-marked ends due to breakage.
10	No centromeres present	Contain centromeres, which contain centromeric DNA and its associated proteins.



(b)

	TRANSCRIPTION	Max 4
1	RNA polymerase II binds at the promoter	1
2	General Transcription Factors Binds at TATA box/ Proximal Control Element → initiates basal rate of transcription	1
3	Activator (specific) Transcription Factors Binds at Enhancer/ Distal Control Element → increases rate of transcription	1
4	Repressor (specific) Transcription Factors Binds at Silencer/ Distal Control Element → Decreases rate of transcription	1
5	Mention of transcription initiation complex/Transcription complex/Enhancesome	1
	POST-TRANSLATIONAL	Max 4
6	Addition of 5' cap for 1. Capping is important in the regulation of export of mRNA from the <u>nucleus</u> . 2. It also helps to prevent <u>degradation</u> by hydrolytic enzymes present in the cell. 3. Aids the attachment of <u>ribosome</u> to the 5' end of the mRNA. 4. ensures the messenger RNA's stability while it undergoes translation in the process of protein synthesis	1
7	Addition of poly-A tail 1. Necessary for transcription termination 2. facilitate export of the mature mRNA from nucleus 3. protect the mRNA from degradation by hydrolytic enzymes poly-A tail help ribosomes attach to the 5' end of the mRNA	1
8	RNA splicing by Small nuclear ribonucleoproteins (snRNPs) recognize the splice sites/ Spliceosome only	1
9	excise introns and exons spliced together	1
	TRANSLATIONAL LEVEL	
10	a. unsuitable mRNA degraded by repression proteins/unfolded mRNA degraded by exosomes b. Blocked by <u>regulatory</u> proteins at mRNA => ribosomes cannot bind to mRNA. c. mRNA lack <u>poly-A</u> tail of sufficient size. d. <u>Translation</u> factors e. Control of <u>degradation</u> of mRNA. f. RNA <u>Interference</u> MicroRNAs → degrade mRNA or prevent expression of genes.	Max 3
	POST-TRANSLATIONAL LEVEL	Max 2
11	Chemical modification for activating/inactivating proteins	1
12	Cleaving to produce functional proteins	1



QUESTION NINE

BIOLOGICAL MOLECULES, CELL MEMBRANE, NERVOUS SYSTEM, ENDOCRINE SYSTEM

(a) Compare a molecule of triglyceride with a molecule of phospholipid. [5]

Similarities:

1. Derived from glycerol;
2. Presence of fatty acid residues;
3. Which may differ in length/ degree of saturation;
4. Presence of ester linkages;

[1/2 mark each]

Differences:

FEATURE	TRIGLYCERIDE	PHOSPHOLIPID
5. No. of Fatty Acid Residues	3	2;
6. Phosphate / Polar Head Group	Absent	Present;
7. Solubility	Hydrophobic/ Insoluble in water	Amphipathic;
8. Function	Metabolic/ Energy reserve/ Respiratory substrate/ Insulation	Structural/ Biological membrane;

[any 3m]

[TOTAL: 5m]



(b) Discuss the properties of a cell membrane that are important in the functioning of a mammalian nerve cell. [10]

Answer:

PROPERTIES OF CELL MEMBRANE	FUNCTIONS OF NERVE CELL
9. Hydrophobic phospholipid core/ Differential permeability (to Na^+ / K^+ /organic ions);	10. Produces/ Restores RP;
11. Na^+ , K^+ ions only able to pass through membrane via transmembrane proteins/ Na^+ - K^+ pump/ ion channels;	
12. Presence of Na^+ - K^+ pump;	
13. Permeability can be altered (opening and closing of Na^+ / K^+ / Ca^{2+} channels);	14. Generation of AP , PSP; 15. Gives rise to refractory period (when closed or inactivated);
16. Permits flow of local current;	17. Propagation of AP;
18. Resealable / Flexible / Fluid;	19. Release of neurotransmitter through exocytosis; 20. Reuptake of neurotransmitter; 21. Growth/ Elongation/ Formation of dendrites;
22. Presence of specific receptors on post-synaptic membrane;	23. Recognise and bind to neurotransmitter; 24. Transmits AP / Sets up PSP; 25. Ensures unidirectionality;
26. Presence of acetylcholinesterase in post-synaptic membrane;	27. Hydrolyses neurotransmitter, hence terminating transmission;
28. Specific receptors in pre-synaptic membrane;	29. Neuronal re-uptake of neurotransmitter;

[TOTAL: 10m (max)]



(c) Suggest reasons why the endocrine system instead of the nervous system is used to control certain physiological processes. [5]

Answer:

- 30. Hormones distributed widely in the blood;
- 31. Advantage of long-lasting control / hormones remain in blood longer;
- 32. Exerts slow effects on named processes like growth / metabolism / development/ reproduction;
- 33. Costly for nerves to have the same effect;
- 34. Fatigue – $\frac{1}{2}$ mark;
- 35. Biological systems generally cannot tolerate sudden fluctuations;
- 36. Restores systems back to normal / Potential for complex interaction among different hormones;
- 37. Easier to maintain homeostasis through negative feedback mechanisms by adjusting hormone levels;

[TOTAL: 5m (max)]

